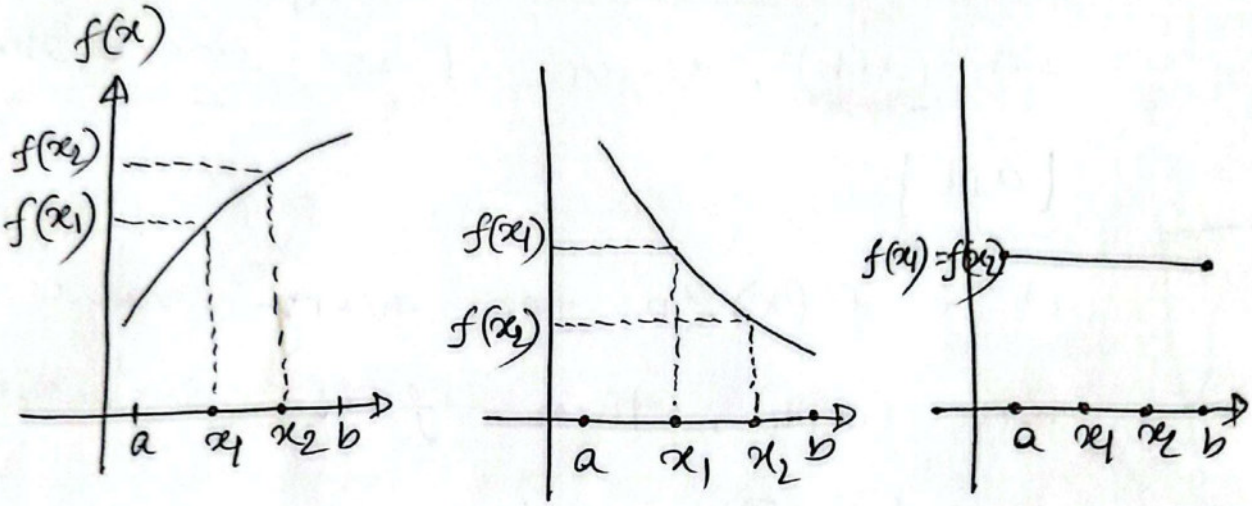
lecture- 05

* Increasing and Decreasing Functions:

→ Let f be defined on an interval and let x_1 and x_2 be two points in that interval.



Increasing Decreasing constant

a) The function $f(x)$ is increasing if $f(x_1) < f(x_2)$ whenever $x_1 < x_2$.

b) The function $f(x)$ is decreasing if $f(x_1) > f(x_2)$ whenever $x_1 < x_2$

c) The function $f(x)$ is constant if $f(x_1) = f(x_2)$ whenever $x_1 < x_2$

CamScanner

* Theorem:

Let f be a function that is continuous on a closed interval on the $[a,b]$, then

a) If $f'(x) > 0$ for every value of x in (a,b), then f is increasing on $[a,b]$

b) If $f'(x) < 0$ for every value of x in (a,b), then f is decreasing on $[a,b]$.

c) If $f'(x) = 0$ for every value of x in (a,b), then f is constant on $[a,b]$.

Example: Find the intervals on which $y = x^3$ is increasing or decreasing.

Ans. Given that $f(x) = y = x^3$

$\Rightarrow f'(x) = 3x^2$

Thus, $f'(x) > 0$ if ~~$x > 0$~~ $x < 0$

& $f'(x) > 0$ if $x > 0$

since f is continuous everywhere, f is increasing on $(-\infty, 0]$ and f is increasing on $[0, \infty)$. therefore f is increasing on $(-\infty, \infty)$. Ans.

<u>Example:</u> Find the intervals on which the function $f(x) = 3x^4 + 4x^3 - 12x^2 + 2$ is increasing or decreasing.

Solu:

Given, $f(x) = 3x^4 + 4x^3 - 12x^2 + 2$

$$f'(x) = 12x^3 + 12x^2 - 12 + 4x$$

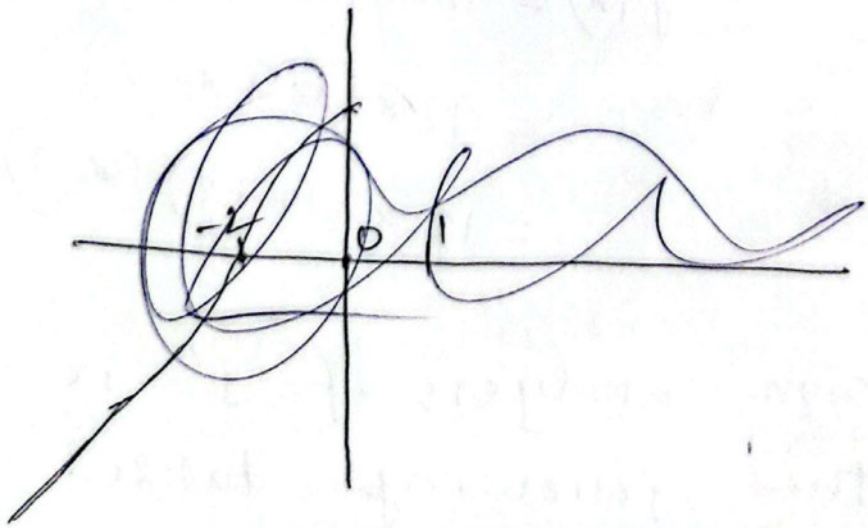
$$= 12x(x^2 + x - 2)$$

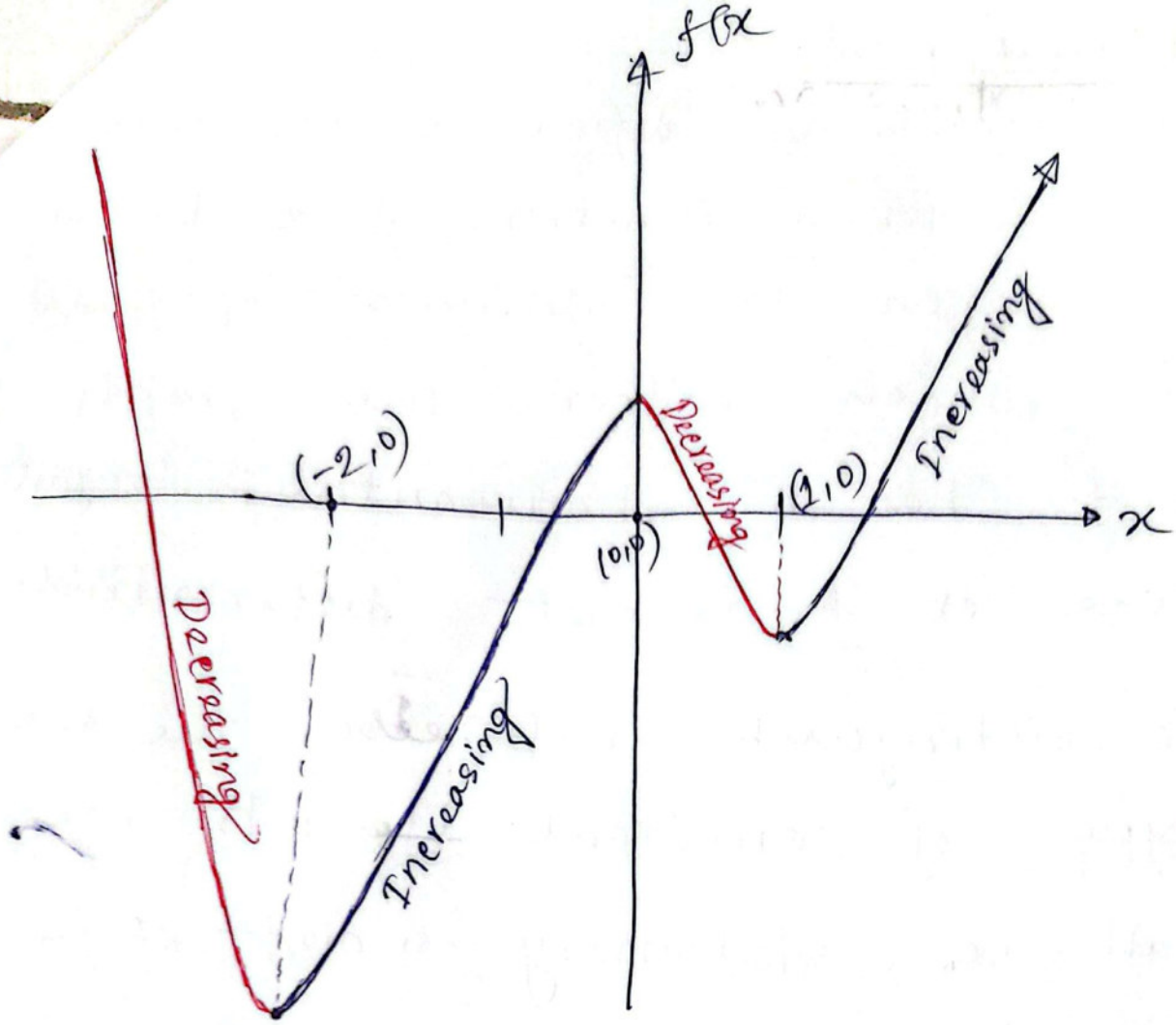
$$= 12x\,(x+2)\,(x-1)$$

The sign analysis of f' is given in the following table.

Interval	$12x(x+2)(x-1)$	$f'(x)$	conclusion
$x < -2$	$(-)(-)(-)$	$-$	Decreasing on $(-\infty, -2]$
$-2 < x < 0$	$(-)(+)(-)$	$+$	Increasing on $[-2, 0]$
$0 < x < 1$	$(+)(+)(-)$	$-$	Decreasing on $[0, 1]$
$x > 1$	$(+)(+)(+)$	$+$	Increasing on $[1, \infty)$

Therefore, the function is Increasing on $(-2, 0)$ & $(1, \infty)$ and decreasing on ~~$(-\infty, 2)$~~ $(-\infty, -2)$ & $(0, 1)$. Ans.





* <u>practise problem</u>:

Exercise 3.5 : 5-10, 11-16

" 10.1 : 53, 54, 45-52

" 4.1 : 15-32 (a,b)

" 4.2 : 7-14, 25-32, 33-50

* critical point:

In general, we define a critical point for a function f to be a point in the domain of f ~~all~~ at which either the graph of f has a horizontal tangent line or f is not differentiable.

To distinguish between the two types of critical points we call x_0 stationary point of f if $f'(x_0) = 0$.

Ex. Find the critical points of ~~f(x)~~

$$f(x) = x^3 - 3x + 1$$

Ans:

$$f'(x) = 3x^2 - 3$$

For critical points $\left\{ \begin{array}{l} f'(x) = 0 \\ \Rightarrow 3x^2 - 3 = 0 \\ \Rightarrow x^2 = 1 \quad \therefore x = \pm 1 \end{array} \right.$